

Is frequent trading profitable or hazardous? Heterogeneity in investor performance*

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Keywords: High-frequency trading; Index futures market; Informed investor; Liquidity provider; Noise trader

JEL classification: G11; G14; G23

* The authors thank Kee H. Chung, Robert Parham, and Robert I. Webb for their valuable comments and suggestions.

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1. Introduction

In this paper, we explore whether high-frequency trading (HFT) is profitable. Specifically, we analyze the relation between trading duration (i.e., an inverse measure of trading frequency) and investment performance and whether this relation varies with investor types, intraday intervals, and market states. Trading speed has risen rapidly following the emergence of automated, decentralized markets over the past several decades (Glosten, 1994; Handa, Schwartz, and Tiwari, 2003; O'Hara, 2015). As Haldane (2011) states, "to be uninformed is to be slow." (p. 6). The continuous and order-driven features of these markets enable HFT because they allow trading with extremely low latency and small transaction costs (Hasbrouck and Saar, 2013).

We measure the trading frequency and performance of each investor using the encrypted account-level transaction data from the highly liquid index futures market in Korea. Because we measure the trading frequency of each investor, our identification of frequent trading is free from errors in the categorization of HFT. We find that domestic non-investment institutions are the most frequent traders, while domestic individuals are the least frequent traders. Other institutional investors' trading frequencies are similar to each other. We also show that trading duration increases as the market approaches the closing time.

Our empirical results suggest the relation between trading duration (or frequency) and performance differs significantly across trader types. Notably, the highest-frequency-trading group (i.e., domestic non-investment institutions) loses money as their trading frequency increases. By contrast, other institutional traders enjoy greater profits when they trade more frequently (i.e., trading duration is shorter). Domestic individuals' investment performance does not depend on their trading frequency. The intraday analysis suggests that the negative relation between domestic non-investment institutions' trading frequencies and performances is pronounced during the market opening hours when liquidity is low. Other institutions'

frequent trading results in greater profits per contract as they approach the market closing time. Finally, we find that domestic non-investment institutions lose money from frequent trading when market liquidity is scarce. Our findings remain robust to the change in market conditions, adjustment for mean price levels, and different model specifications.

The role of HFT differs across the literature. Hendershott, Jones, and Menkveld (2011) suggest that HFT enhances the liquidity of large stocks (e.g., smaller bid-ask spreads and adverse selection costs). Other studies confirm that market liquidity is improved by high-frequency or algorithmic trading (Conrad, Wahal, and Xiang, 2015; Hendershott and Riordan, 2013) and high-frequency traders' role as implicit, endogenous market makers in the automated environment (Brogaard, Carrion, Moyaert, Riordan, Shkilko, and Sokolov, 2018; Carrion, 2013; Menkveld, 2013, 2016). In a similar manner, Brogaard, Hendershott, and Riordan (2014), Chaboud, Chiquoine, Hjalmarsson, and Vega (2014), and Chakrabarty, Moulton, and Wang (2022) show that fast trading improves market efficiency by reducing arbitrage opportunities. Hagströmer and Nördén (2013) and Hasbrouck and Saar (2013) find that short-term volatility decreases as HFT prevails. These studies focus on high-frequency traders' roles to facilitate and stabilize the market.

Despite the bright side of HFT, some studies report contrary conclusions. Easley, De Prado, and O'Hara (2011) claim that high-frequency traders fail to serve as endogenous market makers when they face excessive liquidity dry-ups or shortages, which is represented by the flash crash in 2010. Other studies similarly confirm that these implicit market makers begin to trade *with the wind* when substantial liquidity provision is demanded (Kirilenko, Kyle, Samadi, and Tuzun, 2017; Van Kervel and Menkveld, 2019). Boehmer, Fong, and Wu (2021) also show that algorithm traders' activities improve market liquidity; however, they report that short-term volatility deteriorates following such activities.

Potential adverse effects of HFT are further expressed. High-frequency traders tend to exploit arbitrage opportunities while contributing to market-making roles (Biais and Foucault, 2014; Harris, 2013); however, such immoderate trading may incur side effects. Kozhan and Tham (2012) show that high-frequency traders face a greater risk of carrying out unprofitable arbitrage trades as they compete against each other. The increased execution risk hinders arbitrageurs' participation and deteriorates market efficiency. A group of studies confirms that markets become shallower following excessive low-latency trading activities while their efficiency remains not improved (Budish, Cramton, and Shim, 2015; Foucault, Kozhan, and Tham, 2017; Manahov, 2016). A natural question that arises following the aforementioned findings is whether fast traders devote themselves more to the endogenous market-making role or the arbitrage incentives.

To our best knowledge, this study is the first to explicitly identify high-frequency trading and examine its impacts on investors' performances. Previous studies often employ indirect proxies constructed by observing traders' activities (e.g., Hagströmer and Nordén, 2013), consulting experts (e.g., Benos, Brugler, Hjalmarsson, and Zikes, 2017), borrowing the classification provided by other institutions or exchanges (e.g., Hirschey, 2021), or using the introduction of colocation services (e.g., Brogaard, Hagströmer, Nordén, and Riordan, 2015). We directly measure the average trading duration between consecutive trades for each account and use it as the inverse proxy of its trading frequency. This approach is particularly advantageous because it is free from exterior biases when classifying high-frequency traders. In addition, our results support the existence of distinct, informed liquidity providers in the order-driven context. Unlike the previous convention, we suggest that investors play heterogeneous roles, meaning that some investors serve as liquidity providers while the rest act as speculators or arbitrageurs. The fact that the highest-frequency traders, represented by

domestic non-investment institutions, are in fact the greatest loser from frequent trading implies that these investors are informed liquidity providers.

The remainder of this paper proceeds as follows. Section 2 describes the sample data and methodological approach. Section 3 addresses the preliminary factorization process. Section 4 discusses the empirical findings, and Section 5 checks the robustness of the findings. Section 6 concludes the study.

2. Sample Data and Methodology

2.1. KOSPI 200 index futures market

Our dataset comprises the records of all transactions executed in the Korea Composite Stock Price Index (KOSPI) 200 futures market, from January 4, 2010, to June 30, 2014. We investigate the data from the index futures market for the following reasons. First, this dataset provides in-depth details sufficient to carry out account-level analyses. Each transaction record provides information including quotes and volumes of nearby standing limit orders, encrypted identification of the buyer and seller, and investor-type flags that identify whether they are domestic or not and whether they are individual or institutional. Second, the abundant liquidity of this market and the leveraging effects of futures contracts are attractive to informed traders who intend to speculate on the spot index (i.e., the KOSPI 200 index). Third, the index derivatives are relatively free from idiosyncratic risk and private information asymmetry. Last, distinct types of investors participate in this market. Whereas most emerging markets are strongly dominated by local traders, the account-based participation ratios of individual, domestic institutional, and foreign institutional investors in this market are 43.75%, 29.83%, and 26.42%, respectively.

In this market, there exists a single series of futures contracts per maturity. The maturity dates, or the expiration dates, are the second Thursdays of the last month of each calendar

quarter: March, June, September, and December. All assets in this market expire on maturity dates, and a new series of contracts are traded from the subsequent trading days; thus, the series rolls over every three months. The first two maturity dates in our sample are on March 11 and June 10 in 2010, and the last two are on March 13 and June 12 in 2014.

2.2. Account-level performance

We measure investor performances using the account-level cash flows of purchasing and liquidating futures contracts to investigate the relation between each investor's performance and trading frequency. Because futures contracts are closed out prior to their maturities, our performance measure rolls over on expiration dates. Thus, our measure captures each account's performance during the period from one expiration date to the subsequent one. The detailed measurement procedure is as follows.

First, we sort all transactions by maturity date and by account using the encrypted identification of each trader.

Second, we leave the transactions recorded during the continuous trading session, which is from 9:00 to 15:05.¹ The overnight transactions are excluded because *i*) they use a different quotation limit where the maximum volume for a quote is 100 contracts, while it is 1,000 contracts during the day; *ii*) only limit order trading is allowed; and *iii*) best bid and ask quotes, as well as the standing limit order quantities for each quote, are not disclosed, while they are disclosed up to the fifth-best quotes on both sides during the day. All these differences in trading systems make instantaneous and large-volume trades more challenging during overnight sessions, making information-based trading less likely. The transactions during call

¹ The KOSPI 200 futures market opens at 9:00, after an hour of pre-open call auction session from 8:00 up to 9:00. The market officially closes at 15:15 but the continuous session ends at 15:05; the last ten minutes before the market closing is also a call auction session.

auction sessions are removed because they follow a uniform pricing rule which makes it impossible to measure the trading frequency of each account.

Third, we identify all transactions made by each account. We trace all encrypted identification codes for both the buyer and seller of each transaction and identify all transactions executed by each account. Here, even the limit bid (ask) orders are also counted as an investor's transactions if they are initiated by other investors' ask (bid) orders. Then, using the transaction records, we calculate the profits of each account as its net cash flows from one expiration date to the subsequent expiration date. For example, if an investor purchases a futures contract at \$100 and liquidates it at \$101, then that investor's profit during this period is \$1.

Fourth, we exclude the transaction data before the first maturity (i.e., March 11, 2010) and after the last maturity (i.e., June 12, 2014) because we do not have full records of transactions during these periods.

Fifth, we adjust the account-level profits by the number of contracts traded by each account. The volume-adjusted profit ($Perf_{i,t}$), or the performance measure, is calculated as Equation (1).

$$Perf_{i,t} = \frac{Profit_{i,t}}{Volume_{i,t}}, \quad (1)$$

where $Profit_{i,t}$ denotes the cash flow of account i during the period between the $t+1^{th}$ and t^{th} futures expiration dates in our sample, and $Volume_{i,t}$ is the total number of contracts traded by account i during the corresponding period. The scaling process is necessary because the investors who earn profits of \$1 by trading one futures contract are only as competent as those who earn \$10 by trading ten contracts.

Panel A of Table 1 shows the intraday average performance of each investor type: domestic individuals (*Ind*), domestic investment firms (*Ins_I*), domestic non-investment institutions (*Ins_NI*), foreign investment firms (*For_I*), and foreign non-investment institutions (*For_NI*). Note that the performance measures during certain intraday intervals are negative for all investor types. This is possible because we show the average values of the account-level volume-adjusted profit, not the net cash flow. The negative (positive) average values can be driven by the existence of some investors with extremely large losses (gains).

[Table 1 here]

Foreign investors and domestic investment firms have similar performances throughout the day, exhibiting inverted U-shaped patterns. The average performance is generally similar to each other and remains negative after the market opens until it turns positive after the 11:00-12:00 interval. Domestic non-investment institutions also exhibit similar patterns to other institutions, but they are less profitable shortly after the market opening. Domestic individuals seem rather noisy, and their gains and losses per contract are moderate.² Finally, we find that the overall per-contract profits (shown in the row labeled *Cont. Session*) across all investor types are negative, suggesting that some investors have extremely large losses across all types.

2.3. Trading duration by account

Most studies of intraday trading frequency or trading speed focus on the performances of high-frequency traders restricted to certain institutional traders with HFT flags, often classified

² Here, the value one indicates the profit of one point per contract, and a point is equivalent to 500,000 KRW, which is approximately 395 USD as of 2022. Thus, the average performance of 1 (-1) indicates that the investor gains (loses) 500,000 KRW, on average, per contract.

by experts, exchanges, or other institutions (e.g., Benos, Brugler, Hjalmarsson, and Zikes, 2017; Hirschey, 2021). Others attempt to indirectly identify the existence of high-frequency traders by tracing and observing investors' trading activities (e.g., Hagströmer and Nordén, 2013) or using the introduction of colocation services (e.g., Brogaard, Hagströmer, Nordén, and Riordan, 2015). Taking the advantage of the account-level, tick-by-tick data, we explicitly measure the trading duration of each account and its trading frequency (e.g., Manganelli, 2005). We construct the trading duration measure as follows.

First, we process the data in a similar manner to the way that we constructed the account-level performance measure. We sort all transactions by maturity date and account, using the encrypted identification of each trader. Then, we exclude all transactions during the overnight and call-auction sessions, where only limit order trading is allowed. Because futures contracts are discretely traded, the trading duration or frequency cannot be measured during these sessions. Thus, the trading frequency (duration) measure used in this study specifically indicates how frequently (infrequently) orders are executed by each account during the continuous trading session.

Second, the trading duration is calculated as the distance between two consecutive transactions, as shown in Equation (2). We use the submission time of the order, which is recorded in milliseconds, that initiates each transaction for each account to do so.

$$Time_{i,s} = Timestamp_{i,s} - Timestamp_{i,s-1}, \quad (2)$$

where $Time_{i,s}$ is the time between the s^{th} and $s-1^{th}$ transactions for account i . $Timestamp_{i,s}$ is the timestamp of the s^{th} bid (ask) order submitted by account i when the transaction is buyer-initiated (seller-initiated). Here, a transaction is buyer-initiated (seller-initiated) if the ask (bid)

stamp precedes the bid (ask) stamp. Note that the timestamps are based on the time of order execution, not submission.

Third, we remove all time differences measured across the day. For example, the time difference between the last transaction of an investor on a certain day and the first transaction of that investor on the very next day does not represent trading duration. Thus, all observations at the beginning of the continuous trading sessions are removed to avoid errors from measuring across-the-day differences.

Fourth, we measure the mean trading duration ($Duration_{i,t}$) of each account by calculating the volume-weighted $Time_{i,s}$, which denotes the time from one maturity date to the subsequent one, as shown in Equation (3).

$$Duration_{i,t} = \sum_{s=1}^{S_t} (Volume_{i,s} \cdot Time_{i,s}), \quad (3)$$

where $Volume_{i,s}$ denotes the number of contracts traded at the s^{th} transaction by account i .

Panel B of Table 1 shows the trading duration throughout the day for each investor type. The investors with the shortest duration (or the highest-frequency traders) are domestic non-investment institutions. These investors, on average, trade once every 24 minutes throughout the day, and they trade the most frequently shortly after the market opening by executing a transaction every 4 minutes and 43 seconds. Other types of investors' average duration are all greater than 30 minutes. Not surprisingly, the lowest-frequency traders are domestic individuals. We also note that, regardless of the investor type, the trading duration is the shortest when the market opens and monotonically increases until the market closes, which is consistent with the previous literature (e.g., Admati and Pfleiderer, 1988; Stephan and Whaley, 1990).

2.4. Model specification

The primary goal of this study is to examine the relation between performance and trading duration. Stationarity tests using the augmented Dickey-Fuller method confirm that both trading duration and performance measures are stationary. We use the following regression model to examine the relation between the two variables:

$$Perf_{i,t}^{Inv} = \alpha + \beta \cdot Duration_{i,t}^{Inv} + \gamma' \cdot M_t + \varepsilon_{i,t}^{Inv},$$

where $Inv \in \{Ind, Ins_I, Ins_NI, For_I, For_NI\}$. (4)

Here, $Perf_{i,t}^{Inv}$ denotes the trading performance of investor i , which is calculated as the volume-adjusted net cash flow of account i during the period between the $t+1^{th}$ and t^{th} futures expiration dates in our sample. $Duration_{i,t}$ is the volume-weighted average trading duration of account i during the period between $t+1^{th}$ and t^{th} futures expiration dates. The superscript Inv indicates the investor type. M_t is the set of control variables, measured during the period between the $t+1^{th}$ and t^{th} futures expiration dates, including underlying spot market fundamentals, futures market fundamentals, and macroeconomic variables. Detailed descriptions of the control variables are shown in Table 2. The significantly positive (negative) coefficient β implies that the investors lose (earn) greater profits per contract as they trade more frequently.

[Table 2 here]

3. Preliminary factorization

As a preliminary step before the analysis, we check for multicollinearity issues. The regression results using the model in Equation (4) are shown in Table 3. Models $M1$ through

M6 show the results using different sets of control variables. Model *M1* employs a simple regression model, and *M2* additionally considers the year effect. Model *M3* controls for the return, volatility, and liquidity dynamics in the underlying spot market; *M4* and *M5* control for those in the index futures market. Model *M6* considers the entire set of control variables. The row labeled *Year* shows whether the given model considers the year effect. *F-statistic* shows the *F*-statistic values for model fitness, and *Adj.R²* shows the adjusted *R*-squared values.

[Table 3 here]

The coefficients on trading duration are statistically insignificant as we control for the year effect and remain so in other models. Moreover, the result in model *M6* displays substantially inflated coefficient sizes and smaller *t*-statistics compared to other models. Finally, as shown in the column labeled *VIF* in Table 3, the variance inflation factors are substantially greater than ten, which is the conventional threshold, indicating that multicollinearity is severe in this model.

To avoid the multicollinearity problem without sacrificing the explanatory power of the control variables, we extract major common factors from the control variables using the principal component analysis (PCA) method. The eigenvalue for each factor is shown in Panel A of Table 4. The column labeled *Eigenvalue* shows the eigenvalues of each factor assuming that the sum of all eigenvalues equals the number of control variables (i.e., 16). *N* denotes the order of the factors. *Proportion* shows the proportion of each eigenvalue relative to 16, and *Cumulative* shows the cumulative *Proportion* values considering up to the *Nth* factor. For example, the first component explains about 31% of the total variance. According to Panel A, we find that the cumulative proportion of variance explained increases by about 95%, 99%, and 100% when 8, 11, and 16 principal components, respectively, are considered. We may

derive the same intuition from Fig. 1. After considering 8 principal components, the increment in the cumulative proportion becomes substantially gentle.

[Table 4 here]

[Fig. 1 here]

We extract principal components up to 8, 11, and 16, respectively, so that they account for 95%, 99%, and 100% of the total variance. Panel B of Table 4 confirms that the principal components' variance inflation factors are all one, as shown by columns labeled *Base*. In columns labeled *Duration*, we find that the variance inflation factors do not substantially increase after adding the trading duration variable in each model. The test results imply that the model is relatively free from the multicollinearity problem after the factorization.

Table 5 reports the estimation results again, using the factorized control variables this time. The results remain qualitatively unchanged. Even though the cumulative proportion in Table 4 and Figure 1 implies that it may be sufficient to consider only 8 or 11 principal components, our results show that the model better explains the dependent variable (i.e., investors' performances) as we consider more components. Furthermore, the adjusted *R*-squared value of the model *M6* is 0.0013, which is equivalent to that of *M6* in Table 3, indicating that the explanatory power of the model does not fall after factorization. Because of the highest adjusted *R*-squared value, we mainly interpret the results in model *M6* throughout the paper.

[Table 5 here]

Despite the factorization, we can still interpret the relationships between investor performance and some market control variables. Table 6 reports the correlations between the set of control variables and the extracted components. Here, the darker the color of a cell, the higher the correlation. According to the table, *Prin1* represents the growth in futures and spot markets as it is highly correlated with *Ret_Spot*, *Ret_Fu*, and *Cap_Spot*. *Prin2* represents market volatility as its correlation with *Volat_Spot* and *Volat_Fu* are both greater than 0.95. *Prin3* and *Prin8* are associated with the growth and level of market depth, respectively, both representing futures market liquidity. *Prin4*, *Prin5*, and *Prin6* are mostly associated with the growth rates of the credit spread, term spread, and risk-free rate, respectively. *Prin9* is strongly correlated with the exchange rate, and *Prin10* is mostly associated with the changes in futures market volatility.

[Table 6 here]

4. Empirical findings

4.1. Analysis by investor type

In this section, we carry out the subgroup analysis for each investor type. Our dataset provides investor-type flags. The first flag reveals whether an investor is domestic or not; the second reveals whether an investor is an individual, an investment firm, or a non-investment institution. Using the combination of these flags, we categorize investors into five distinct groups: domestic individuals, domestic investment firms, domestic non-investment institutions, foreign investment firms, and foreign non-investment institutions. As previously stated, we

focus on the results using the model that considers all 16 principal components and the year effect.³

Table 7 reports the relationship between trading duration and performance for each investor type after controlling for variables in the spot and futures markets. The estimated coefficients of the principal components imply the following. *Prin1* is positively associated with the dependent variable across all types of investors, indicating that investors generally earn more when the growth rates of futures and spot markets are higher. The coefficients of *Prin2* imply that the performances of Ind and Ins_I significantly deteriorate as the markets become more volatile. The coefficients of *Prin3* and *Prin8* consistently imply that abundant depth in the futures market only benefits domestic investors. The insignificant relationships between these components and foreign investors' performances may be because foreign investors often submit small-sized orders which do not require a large number of standing limit orders (Chou and Wang, 2009; Ryu and Yu, 2021).

[Table 7 here]

The first column, which is labeled *Ind*, shows the result of the subsample analysis for domestic individuals. We find that domestic individuals' profits from trading do not depend on how quickly they trade. Previous studies generally suggest that individual investors are largely noisy and uninformed (e.g., Kostopoulos, Meyer, and Uhr, 2020; Kumar and Lee, 2006), and such claims are often based on these investors' poor performance (e.g., Barber, Odean, and Zhu, 2009) and lack of trading skills (e.g., Hu, Kirilova, Park, and Ryu, *forthcoming*). The finding that their higher-frequency trading is just as profitable as lower-frequency trading is

³ We confirm that all results are qualitatively unchanged when considering different numbers of principal components (e.g., 8 and 11) and without the year effect.

somewhat consistent with conventional wisdom. On the other hand, unlike the well-known notion claimed by Barber and Odean (2000), it also suggests that frequent trading is not necessarily hazardous to individual investors in this market.

Another notable finding is that domestic non-investment institutions do not gain profits from frequent trading. In fact, they lose more as their trading frequency increases. The significantly positive coefficient of *Duration* indicates that their performances improve as the duration increases. The estimated coefficient (i.e., 0.00067) suggests that they earn 335 KRW less per contract when their average trading duration decreases by a second. The loss is calculated per contract and is quite substantial considering that the average trading volume during each analysis window (i.e., from an expiration date to the subsequent one) is about 578; they lose additional 193,631 KRW (or 151.45 USD) every time their trading duration increases by one second. This is strictly against the traditional view in the HFT world that higher-frequency traders earn greater profits (e.g., Budish, Cramton, and Shim, 2015; Manahov, 2016), especially when they are the highest-frequency traders in this market (see Panel B of Table 1). These results may be attributed either to excessive arbitrage-seeking activities (Kozhan and Tham, 2012) or informed liquidity provision of these domestic institutions (Hendershott and Riordan, 2013; Menkveld, 2016).

Lastly, the results for other institutions, including domestic investment firms, foreign investment firms, and foreign non-investment institutions, are somewhat similar to each other and comply with the existing literature: the more frequently they trade, the greater profits they earn. The negative coefficients indicate that their performances improve as they trade more frequently, meaning that they are informed and gain profits from high-frequency trading. This tendency is the most significant and pronounced for foreign investment firms among others. The average trading volumes during each analysis window of domestic investment firms, foreign investment firms, and foreign non-investment institutions are 2231.99, 2114.88, and

2910.03, respectively. Assuming that the relationship is linear, the estimated coefficients for *Ins_I*, *For_I*, and *For_NI* indicate that these investors additionally gain 145,079, 444,125, and 261,903 KRW (or 113.48, 347.38, and 204.85 USD), respectively, whenever their duration decreases by a second. These findings confirm that foreigners are the most directionally informed traders (Froot, O'Connell, and Seasholes, 2001; Richards, 2005; Ryu, 2015) and gain excess profits from trading in emerging markets (Barber, Lee, Liu, and Odean, 2009; Bekaert and Harvey, 2000). In addition, we suggest that foreign investment firms are the most informed even compared to other foreign institutions and that domestic investment firms also trade with information.

4.2. Intraday analysis

Tables 1 and 2 suggest that investors' trading patterns vary not only with their types but also throughout the day. In this section, we investigate whether the relationship between trading duration and performance is affected by the time of the day. The regression results are shown in Table 8. We primarily note that domestic non-investment institutions' trading frequency significantly and negatively affects their profits only during the market opening period (i.e., from 9:00 to 10:00). Knowing that their trading duration is the shortest in the morning, such frequent trading appears to be intentional because excessive competition in arbitrage-seeking activities may incur market inefficiency but not economic losses for individual accounts. Simply put, these investors can save their money simply by turning away from the market during this period. We believe that this partially supports the liquidity-providing role of domestic non-investment institutions, considering that the market is the most illiquid shortly after it opens (Upson and Van Ness, 2017).

[Table 8 here]

Other institutions' frequent trading is, again, more straightforward. Their higher-frequency trading yields greater profits, and such tendencies generally become more apparent as they approach the market closing. In other words, the aforementioned relationships become stronger as market liquidity improves throughout the day. Because traders with information purportedly time their transactions in order to trade behind abundant liquidity (Chakravarty, 2001; Collin-Dufresne and Fos, 2015, 2016), we may interpret that these investors are informed institutional traders who carry out high-frequency trading in the pursuit of economic profits.

4.3. Relationship between duration and performance depending on liquidity states

We further investigate whether the impact of frequent trading on investors' per-contract profits varies with market liquidity states. As we have addressed, arbitrageurs or informed speculators tend to participate more when liquidity is ample. Contrarily, if the negative relationship between trading duration and performance is owing to their commitment to market-making roles, then such a relationship will be pronounced when liquidity is scarce. To test this hypothesis, we measure the volume-weighted averages of the bid-ask spread and market depth⁴, respectively, for each analysis window, from one maturity date to the subsequent one. The periods with the top (bottom) 30 percentiles of the bid-ask spread (market depth) are categorized as low liquidity periods; those with the bottom (top) 30 percentiles of the bid-ask spread (market depth) are categorized as high liquidity periods.

Table 9 shows the test results. We find that domestic non-investment institutions' frequent trading is positively related to trading losses only when liquidity is scarce. This result is in favor of our conjecture that these investors' high-frequency trading aims to provide liquidity.

⁴ We calculate the market depth as the sum of all standing limit orders up to the fifth-best bid and ask quotes.

On the other hand, that of foreign investors is generally more profitable during low liquidity periods; however, our results do not discuss whether informed trading causes low liquidity or vice versa.

[Table 9 here]

4.4. Relationship between duration and performance by duration level

Thus far, the overall results indicate that the relationship between trading duration and investment performance depends on the type of investors. Such results may be endogenously driven by the average trading duration of each investor type. For example, this relationship for the *Ins_NI* subsample is positive because they consist of high-frequency traders, on average, not because they are non-investment institutions. In order to test this hypothesis, this section investigates how the relationship varies with the quintile of trading duration, regardless of the investor type.

Panel A of Table 10 shows the composition of investors, in percentage terms, in each quintile of trading duration. *Q1* (*Q5*) indicates the quintile with the lowest (highest) trading duration. Notably, in terms of the number of accounts, domestic individual investors take the largest proportion in all quintiles. Their proportion is the highest (i.e., 46.4%) in the bottom quintile, *Q5*, denoting they generally trade with relatively low frequencies. It monotonically decreases as the quintile approaches *Q1*. Contrarily, domestic non-investment institutions' proportions are less than 5% in lower quintiles (i.e., *Q3*, *Q4*, and *Q5*). It reaches 6.62% in *Q2* and 17.43% in *Q3*. This indicates that domestic non-investment institutions are generally high-frequency traders. Other groups, on the other hand, exhibit similar levels of proportion through all quintiles.

[Table 10 here]

The impact of trading duration on investors' profits for each quintile of trading duration is shown in Panel B of Table 10. We find that the coefficient of *Duration* is only significant and positive for the quintile with the lowest trading duration. Because the most apparent differences between the top quintile, *Q1*, and the other quintiles are the proportions of domestic individuals and domestic non-investment institutions, according to Panel A, the positive relationship is likely attributed to the increased proportion of domestic non-investment institutions. In other quintiles, the relationships are insignificant. The result in this table suggests that the relationship depends on the types of investors, rather than the trading duration itself.

5. Robustness checks

5.1. Biases from the selection of estimation windows

A major concern for the robustness of our findings relates to the biases following the selection of duration periods. Suppose the price of an asset (i.e., index futures) monotonically increases (decreases) over time. All other things being equal, investors who trade this asset more frequently and execute more round-trip transactions may experience relatively smaller economic profits (losses) per transaction than those who trade less frequently. In other words, the calculated performance of a given investor may be positively correlated with the measurement interval (or the length of the analysis window), providing misleading interpretations.

If such a mechanical relationship exists in our dataset, then traders who trade less frequently or have longer duration should earn greater (smaller) profits on average during the period when futures price rises (falls). Thus, we investigate the subsample statistics to check

whether such a relationship holds in our dataset. Panel A of Table 11 shows the mean and standard deviation values of the trading duration of each investor type in bull and bear markets. We identify the rate of the return from the underlying index during each expiration-to-expiration period and those with the highest and lowest returns as bull-market (*Bull*) and bear-market (*Bear*) periods. The row labeled *Pct15* (*Pct30*) shows the statistics when market states are determined by the top and bottom 15 (30) percentiles.

[Table 11 here]

In Table 11, we observe that domestic non-investment institutions have the lowest trading duration on average and that domestic individuals have the longest duration. These relationships hold for both market states. All traders seem to trade faster in a bull market, and such a tendency is pronounced for foreign investment firms who increased their trading duration by 11.22% when they trade in a bear market. *Duration* of *Ins_NI* is the least dependent on the market state as it only increases by 0.30% in a bear market. If the aforementioned mechanism exists, then the performance of *Ins_NI* should experience cyclicity the least because they are the highest-frequency traders in this market, whereas the performance of *Ind* should be the most cyclical as they are the least frequent traders.

Contrary to the conjecture, we observe, in Panel B of Table 11, that investors' mean trading duration is negatively associated with the cyclicity of their performances. Panel B primarily shows that account-level investment performances depend on the market state, on average, regardless of the types of investors. While all types are subject to cyclicity, domestic individuals, who are the lowest-frequency traders, are affected the least by the market state. Conversely, the performances of domestic non-investment institutions exhibit the most cyclical behavior. The statistics indicate that the concerned mechanical correlation is not observed in

our dataset, and this tendency remains largely unchanged even if different standards (i.e., *Pct15* and *Pct30*) are used.

5.2. Price-adjusted investor performance

In this section, we check whether our empirical findings remain robust to the change in the investor performance measure. Owing to the construction of *Perf*, investors' performances in a given period may be exaggerated when the level of the corresponding futures price is relatively high. Conversely, they may be underestimated when the price is relatively low. We control for the biases by adjusting the performance measure for the mean price in each window.

Table 12 shows the relationship between the trading duration and price-adjusted investor performance by investor type. We find that the relationships remain qualitatively unchanged even after controlling for the effect of the futures price level. The performances of *Ins_I*, *For_I*, and *For_NI* improve as they trade more frequently, and only *Ins_NI* loses money as their trading duration decreases. We consistently find that the performance of *Ind* is not associated with their trading frequency. Furthermore, the estimated coefficients of the control variables are qualitatively similar to those in Table 7, implying that, overall, the level of the futures price does not substantially influence our findings.

[Table 12 here]

5.3. Fama-MacBeth regression

Finally, we test whether our results are robust to the change in methodological approaches. In order to consistently estimate the results, we also employ Fama-MacBeth regression models to consider the potential existence of cross-sectional correlations. Table 13 shows the regression results, and Panel A (Panel B) shows the results estimated based on yearly

(expiration-to-expiration) windows, respectively. Newey-West standard errors are considered. Here, all control variables are not estimated because of near or perfect multicollinearity as they are constructed on the analysis window (i.e., expiration-to-expiration).

[Table 13 here]

The overall results remain qualitatively unchanged in Table 13. We find that the duration and performances of domestic (foreign) non-investment institutions are positively (negatively) related, while those of *Ind* remain insignificant. Although the relationships for investment firms (i.e., *Ins_I* and *For_I*) appear statistically insignificant, the sign of their coefficients remains negative.

6. Conclusion

We investigate the relationship between investors' trading frequencies and performances on account levels. The answer to the question is negative. We find that the act of frequent trading heterogeneously affects investors' performances depending on their types. Notably, the highest-frequency trading party, represented by domestic non-investment institutions, significantly loses money as their trading duration decreases. Contrarily, other institutional traders gain greater profits per contract as their trading duration becomes shorter. Domestic individuals are the lowest-frequency traders and experience neither additional gains nor losses from more frequent trading. In addition, the negative relationship between domestic non-investment institutions' trading frequencies and investment performances is only significant when the market opens and when liquidity is scarce. Other institutions' higher-frequency trading results in better performances as they approach the market closing.

This study is the very first attempt to explicitly examine the impact of high-frequency trading on investors' performances by measuring the trading duration of each account. Various measures of HFT are addressed by previous studies, but they only indirectly proxy its existence. Our approach is advantageous because it does not rely on classification standards. Moreover, we provide partial evidence of informed liquidity providers in the order-driven market. Our findings are somewhat in contrast to previous studies in that different types of investors distinctly commit to the roles of arbitrageurs and liquidity providers.

References

- Admati, A.R., & Pfleiderer, P. (1988). A theory of intraday patterns: Volume and price variability. *Review of Financial Studies*, 1(1), 3-40.
- Barber, B.M., Lee, Y.-T., Liu, Y.-J., & Odean, T. (2009). Just how much do individual investors lose by trading? *Review of Financial Studies*, 22(2), 609-632.
- Barber, B.M., & Odean, T. (2000). Trading is hazardous to your wealth: The common stock investment performance of individual investors. *Journal of Finance*, 55(2), 773-806.
- Barber, B.M., Odean, T., & Zhu, N. (2009). Systematic noise. *Journal of Financial Markets*, 12(4), 547-569.
- Bekaert, G., & Harvey, C. R. (2000). Foreign speculators and emerging equity markets. *Journal of Finance*, 55(2), 565-613.
- Benos, E., Brugler, J., Hjalmarsson, E., & Zikes, F. (2017). Interactions among high-frequency traders. *Journal of Financial and Quantitative Analysis*, 52(4), 1375-1402.
- Biais, B., & Foucault, T. (2014). HFT and market quality. *Bankers, Markets & Investors*, 128, 5-19.
- Boehmer, E., Fong, K., & Wu, J.J. (2021). Algorithmic trading and market quality: International evidence. *Journal of Financial and Quantitative Analysis*, 56(8), 2659-2688.
- Brogaard, J., Carrion, A., Moyaert, T., Riordan, R., Shkilko, A., & Sokolov, K. (2018). High frequency trading and extreme price movements. *Journal of Financial Economics*, 128(2), 253-265.
- Brogaard, J., Hagströmer, B., Nordén, L., & Riordan, R. (2015). Trading fast and slow: Colocation and liquidity. *Review of Financial Studies*, 28(12), 3407-3443.
- Brogaard, J., Hendershott, T., & Riordan, R. (2014). High-frequency trading and price discovery. *Review of Financial Studies*, 27(8), 2267-2306.

- Budish, E., Cramton, P., & Shim, J. (2015). The high-frequency trading arms race: Frequent batch auctions as a market design response. *Quarterly Journal of Economics*, 130(4), 1547-1621.
- Carrion, A. (2013). Very fast money: High-frequency trading on the NASDAQ. *Journal of Financial Markets*, 16(4), 680-711.
- Chaboud, A.P., Chiquoine, B., Hjalmarsson, E., & Vega, C. (2014). Rise of the machines: Algorithmic trading in the foreign exchange market. *Journal of Finance*, 69(5), 2045-2084.
- Chakravarty, S. (2001). Stealth-trading: Which traders' trades move stock prices? *Journal of Financial Economics*, 61(2), 289-307.
- Chakrabarty, B., Moulton, P.C., & Wang, X.F. (2022). Attention: How high-frequency trading improves price efficiency following earnings announcements. *Journal of Financial Markets*, 57, 100690.
- Chou, R.K., & Wang, Y.-Y. (2009). Strategic order splitting, order choice, and aggressiveness: Evidence from the Taiwan futures exchange. *Journal of Futures Markets*, 29(12), 1102-1129.
- Collin-Dufresne, P., & Fos, V. (2015). Do prices reveal the presence of informed trading? *Journal of Finance*, 70(4), 1555-1582.
- Collin-Dufresne, P., & Fos, V. (2016). Insider trading, stochastic liquidity, and equilibrium prices. *Econometrica*, 84(4), 1441-1475.
- Conrad, J., Wahal, S., & Xiang, J. (2015). High-frequency quoting, trading, and the efficiency of prices. *Journal of Financial Economics*, 116(2), 271-291.
- Easley, D., De Prado, M.M.L., & O'Hara, M. (2011). The microstructure of the "flash crash": Flow toxicity, liquidity crashes, and the probability of informed trading. *Journal of Portfolio Management*, 37(2), 118-128.
- Foucault, T., Kozhan, R., & Tham, W.W. (2017). Toxic arbitrage. *Review of Financial Studies*, 30(4), 1053-1094.
- Froot, K.A., O'Connell, P.G.J., & Seasholes, M.S. (2001). The portfolio flows of international investors. *Journal of Financial Economics*, 59(2), 151-193.
- Glosten, L.R. (1994). Is the electronic open limit order book inevitable? *Journal of Finance*, 49(4), 1127-1161.
- Hagströmer, B., & Nordén, L. (2013). The diversity of high-frequency traders. *Journal of Financial Markets*, 16(4), 741-770.
- Haldane, A. (2011, July 8). *The race to zero*. International Economic Association Sixteenth World Congress, Beijing, China. <https://www.bankofengland.co.uk/speech/2011/the-race-to-zero-speech-by-andy-haldane>
- Handa, P., Schwartz, R., & Tiwari, A. (2003). Quote setting and price formation in an order driven market. *Journal of Financial Markets*, 6(4), 461-489.

- Harris, L. (2013). What to do about high-frequency trading. *Financial Analysts Journal*, 69(2), 6-9.
- Hasbrouck, J., & Saar, G. (2013). Low-latency trading. *Journal of Financial Markets*, 16(4), 646-679.
- Hendershott, T., Jones, C.M., & Menkveld, A.J. (2011). Does algorithmic trading improve liquidity? *Journal of Finance*, 66(1), 1-33.
- Hendershott, T., & Riordan, R. (2013). Algorithmic trading and the market for liquidity. *Journal of Financial and Quantitative Analysis*, 48(4), 1001-1024.
- Hirschey, N. (2021). Do high-frequency traders anticipate buying and selling pressure? *Management Science*, 67(6), 3321-3345.
- Hu, J., Kirilova, A., Park, S.G., Ryu, D. (Forthcoming). Who profits from trading options? *Management Science*.
- Kirilenko, A., Kyle, A.S., Samadi, M., & Tuzun, T. (2017). The flash crash: High-frequency trading in an electronic market. *Journal of Finance*, 72(3), 967-998.
- Kostopoulos, D., Meyer, S., & Uhr, C. (2020). Google search volume and individual investor trading. *Journal of Financial Markets*, 49, 100544.
- Kozhan, R., & Tham, W.W. (2012). Execution risk in high-frequency arbitrage. *Management Science*, 58(11), 2131-2149.
- Kumar, A., & Lee, C.M. (2006). Retail investor sentiment and return comovements. *Journal of Finance*, 61(5), 2451-2486.
- Manahov, V. (2016). A note on the relationship between high-frequency trading and latency arbitrage. *International Review of Financial Analysis*, 47, 281-296.
- Manganelli, S. (2005). Duration, volume and volatility impact of trades. *Journal of Financial Markets*, 8(4), 377-399.
- Menkveld, A.J. (2013). High frequency trading and the new market makers. *Journal of Financial Markets*, 16(4), 712-740.
- Menkveld, A.J. (2016). The economics of high-frequency trading: Taking stock. *Annual Review of Financial Economics*, 8, 1-24.
- O'Hara, M. (2015). High frequency market microstructure. *Journal of Financial Economics*, 116(2), 257-270.
- Richards, A. (2005). Big fish in small ponds: The trading behavior and price impact of foreign investors in Asian emerging equity markets. *Journal of Financial and Quantitative Analysis*, 40(1), 1-27.
- Ryu, D. (2015). The information content of trades: An analysis of KOSPI 200 index derivatives. *Journal of Futures Markets*, 35(3), 201-221.
- Ryu, D., & Yu, J. (2021). Informed options trading around holidays. *Journal of Futures Markets*, 41(5), 658-685.

- Stephan, J.A., & Whaley, R.E. (1990). Intraday price change and trading volume relations in the stock and stock option markets. *Journal of Finance*, 45(1), 191-220.
- Upton, J., & Van Ness, R.A. (2017). Multiple markets, algorithmic trading, and market liquidity. *Journal of Financial Markets*, 32, 49-68.
- Van Kervel, V., & Menkveld, A.J. (2019). High-frequency trading around large institutional orders. *Journal of Finance*, 74(3), 1091-1137.

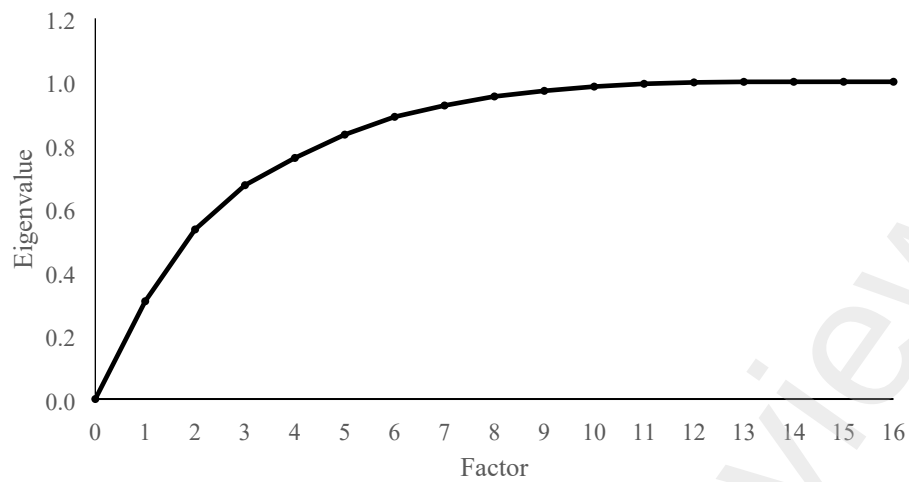


Figure 1. Cumulative proportions of variance explained

Notes: This figure shows the cumulative proportions of variance explained depending on the number of factors considered. The horizontal axis labeled *Factor* shows the number of factors; the vertical axis labeled *Eigenvalue* shows the cumulative proportion of eigenvalues.

Table 1

Descriptive statistics

Panel A. Account-level performance by investor type and intraday time interval

	Ind	Ins_I	Ins_NI	For_I	For_NI
09:00-10:00	-0.1012	-2.1047	-16.5955	-1.3569	-1.5083
10:00-11:00	-0.8155	-3.7162	-4.0326	-3.6548	-2.7571
11:00-12:00	0.1554	1.7373	2.6206	0.3950	0.9908
12:00-13:00	-0.0024	2.2263	8.2556	2.2950	2.3887
13:00-14:00	0.2525	2.7401	10.3841	2.0326	1.1505
14:00-15:00	0.4606	0.4431	8.8223	-0.8909	0.1471
Cont. Session	-0.4987	-0.4413	-1.4729	-1.3799	-0.3910

Panel B. Account-level trade duration by investor type and intraday interval

	Ind	Ins_I	Ins_NI	For_I	For_NI
09:00-10:00	7' 55"	7' 27"	4' 43"	7' 21"	7' 24"
10:00-11:00	21' 25"	19' 35"	15' 5"	19' 38"	19' 17"
11:00-12:00	32' 4"	29' 6"	23' 36"	29' 22"	28' 51"
12:00-13:00	44' 59"	41' 2"	35' 37"	40' 57"	40' 32"
13:00-14:00	54' 14"	48' 17"	41' 51"	48' 58"	48' 18"
14:00-15:00	61' 11"	53' 5"	43' 39"	54' 4"	52' 49"
Cont. Session	36' 55"	33' 8"	24' 21"	32' 27"	32' 46"

Notes: This table reports the descriptive statistics of investors' performances and trading duration by investor type and intraday interval. Panel A (Panel B) shows the mean values of the average account-level performance (duration). *Ind*, *Ins_I*, *Ins_NI*, *For_I*, and *For_NI* indicate domestic individuals, domestic investment firms, domestic non-investment institutions, foreign investment firms, and foreign non-investment institutions, respectively. The labels on rows indicate the intraday time intervals. *Cont. Session* shows the average duration of transactions during the entire continuous trading session by investor type.

Table 2

Variable descriptions

Variable	Description
Perf	Net cash flow per account obtained during the continuous trading session.
Duration	Average trading duration in seconds per account.
Spot market variables	
Ret_Spot	KOSPI 200 index rate of return.
Cap_Spot	The growth rate of the KOSPI 200 market capitalization.
Volat_Spot	The standard deviation of KOSPI 200 index.
Volat_Spot_r	VKOSPI rate of return.
Volume_Spot	Average trading volume in the KOSPI 200 market.
Futures market variables	
Ret_Fu	KOSPI 200 index futures rate of return.
Volat_Fu	Price standard deviation of KOSPI 200 futures.
Volat_Fu_r	Rate of return for intraday price volatility.
Liq_Spr	Volume-weighted average bid-ask spread.
Liq_Dep	Volume-weighted average market depth.
Liq_Spr_r	The growth rate of the bid-ask spread.
Liq_Dep_r	The growth rate of the market depth.
Macroeconomic variables	
Term_Spr_r	The growth rate of the spread between the yields from 10-year government bonds and 2-year monetary stabilization bonds.
Credit_Spr_r	The growth rate of the spread between the yields from 3-year corporate bonds rated AA- and BBB-.
RF_r	The growth rate of the risk-free rate (i.e., yields from 91-day certificates of deposit).
Ex_Rate_r	The growth rate of the USD-KRW exchange rate.

Notes: This table provides definitions and descriptions of all variables used in this study.

Table 3

Preliminary test for multicollinearity

Panel A. Estimating the impact trading frequency on investors' performance with control variables

	M1	M2	M3	M4	M5	M6	VIF
Intercept	-0.4602*** (-4.42)	0.0266 (0.15)	-1.4577*** (-4.86)	-37.6908** (-2.07)	-14.2559** (-2.48)	647.9066 (0.02)	
Duration	-0.0001* (-1.77)	-0.0001 (-1.37)	-0.0000 (-1.15)	-0.0000 (-1.12)	-0.0000 (-1.22)	-0.0000 (-1.24)	1.01
Ret_Spot			-7.3314 (-0.40)			-175.2508 (0.00)	905
Ret_Fu				13.2277*** (7.31)	16.8756*** (9.82)	151.2812 (0.00)	540
Cap_Spot			16.6779 (0.91)			-57.7005 (0.00)	1,514
Volat_Spot			0.3413*** (9.13)			26.9492 (0.01)	16,441
Volat_Fu				0.0233* (1.83)	0.0421*** (5.59)	-4.8717 (-0.01)	13,604
Volat_Spot_r			-3.8890*** (-6.41)			-22.0508 (-0.06)	139
Volat_Fu_r				0.4671*** (2.97)	0.1928 (1.33)	-1.7958 (0.00)	28.11
Volume_Spot			-2.7579*** (-5.23)			-9.9885 (0.00)	5.51
Liq_Spr				715.4186** (1.96)		-11260.000 (-0.06)	138
Liq_Spr_r				-30.7298*** (-4.42)		-10.5175 (0.00)	18.03
Liq_Dep					1.4642** (2.05)	175.3758 (0.00)	25.73
Liq_Dep_r					1.3867*** (6.89)	2.8766 (0.23)	7.74
Term_Spr_r						2.6678 (0.00)	32.96
Credit_Spr_r						-9.7764 (0.00)	7.93
RF_r						36.2696 (0.02)	50.03
Ex_Rate_r						47.7902 (0.03)	30.12
Year	No	Yes	Yes	Yes	Yes	Yes	
F-statistic	3.14*	16.61***	29.02***	26.64***	27.90***	20.67***	
Adj.R²	0.0000	0.0002	0.0009	0.0008	0.0009	0.0013	

Notes: This table shows the relationship between trading duration and performance on account levels. The dependent variable is the account-level net cash flow per contract. *Duration* denotes the account-level trade duration. *Ret_Spot*, *Cap_Spot*, *Volat_Spot*, *Volat_Spot_r*, and *Volume_Spot* denote the rate of return, growth rate of the market capitalization, equity index standard deviation, VKOSPI rate of return, and trading volume, respectively, in the KOSPI 200 market. *Ret_Fu*, *Volat_Fu*, *Volat_Fu_r*, *Liq_Spr*, *Liq_Dep*, *Liq_Spr_r*, and *Liq_Dep_r* denote the rate of return, price standard deviation, rate of return for intraday price volatility, bid-ask spread, market depth, growth rate of the bid-ask spread, and growth rate of the market depth, respectively, in the index futures market. *Term_Spr_r*, *Credit_Spr_r*, *RF_r*, and *Ex_Rate_r* denote the growth rates of the term spread, credit spread, risk-free rate, and USD-KRW exchange rate, respectively. The numbers in parentheses are the *t*-statistics. The column labeled *VIF* shows the variance inflation factors for variables used in model *M6*. The row labeled *Year* shows whether the model controls for the year effect. *F-statistic* shows the *F*-statistic value

for model fitness. $Adj.R^2$ denotes the adjusted R -squared value. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 4

Factorization using the principal component analysis

Panel A. Eigenvalues of the correlation among all variables

N	Eigenvalue	Proportion	Cumulative
1	4.9309	0.3082	0.3082
2	3.6213	0.2263	0.5345
3	2.2320	0.1395	0.6740
4	1.3736	0.0859	0.7599
5	1.1766	0.0735	0.8334
6	0.8969	0.0561	0.8895
7	0.5719	0.0357	0.9252
8	0.4555	0.0285	0.9537
9	0.2834	0.0177	0.9714
10	0.2144	0.0134	0.9848
11	0.1402	0.0088	0.9936
12	0.0708	0.0044	0.9980
13	0.0288	0.0018	0.9998
14	0.0023	0.0001	0.9999
15	0.0012	0.0001	1.0000
16	0.0000	0.0000	1.0000

Panel B. Variance inflation factors after factorization

	8-Var		11-Var		16-Var	
	Base	Duration	Base	Duration	Base	Duration
Duration		1.0046		1.0051		1.0056
Prin1	1.0000	1.0001	1.0000	1.0002	1.0000	1.0002
Prin2	1.0000	1.0037	1.0000	1.0032	1.0000	1.0035
Prin3	1.0000	1.0001	1.0000	1.0003	1.0000	1.0005
Prin4	1.0000	1.0004	1.0000	1.0001	1.0000	1.0003
Prin5	1.0000	1.0003	1.0000	1.0003	1.0000	1.0004
Prin6	1.0000	1.0001	1.0000	1.0003	1.0000	1.0001
Prin7	1.0000	1.0002	1.0000	1.0002	1.0000	1.0002
Prin8	1.0000	1.0001	1.0000	1.0004	1.0000	1.0004
Prin9			1.0000	1.0002	1.0000	1.0002
Prin10			1.0000	1.0001	1.0000	1.0003
Prin11			1.0000	1.0008	1.0000	1.0008
Prin12					1.0000	1.0001
Prin13					1.0000	1.0006
Prin14					1.0000	1.0002
Prin15					1.0000	1.0004
Prin16					1.0000	1.0005

Notes: This table shows the factorization process of control variables. Panel A shows the eigenvalues of principal components extracted using the principal component analysis (PCA) method. *Eigenvalue* shows the eigenvalue of each factor assuming that the sum of all eigenvalues equals the number of control variables. *N* denotes the order of the factors. *Proportion* shows the proportion of each eigenvalue relative to 16, and *Cumulative* shows the cumulative proportion of variance explained, using up to the *N*th factor. Panel B shows the variance inflation factor (VIF) after factorization. The columns labeled *Base* (*Duration*) show the VIFs of principal components without (with) the duration variable. *8-Var*, *11-Var*, and *16-Var* indicate that 8, 11, and 16 principal components, respectively, are extracted using the PCA method. *Duration* denotes trading duration. *Prin1* through *Prin16* denotes the first through sixteenth principal components.

Table 5

Impact of trading frequency on investors' performance on account levels with factorization

	M1	M2	M3	M4	M5	M6
Intercept	-0.49388*** (-4.74)	3.23747*** (7.32)	-0.49668*** (-4.77)	-0.88904 (-0.72)	-0.49945*** (-4.79)	-4.60763*** (-4.87)
Duration	-0.00004 (-1.29)	-0.00004 (-1.17)	-0.00004 (-1.27)	-0.00004 (-1.21)	-0.00004 (-1.23)	-0.00004 (-1.24)
Prin1	0.94133*** (12.08)	0.65213*** (7.53)	0.87398*** (11.22)	0.99618*** (7.83)	0.87031*** (11.17)	0.86215*** (10.99)
Prin2	0.53617*** (6.89)	1.59732*** (8.26)	0.70825*** (9.10)	0.69396** (2.06)	0.68828*** (8.84)	-2.64244*** (-3.66)
Prin3	0.41049*** (5.27)	0.87859*** (7.94)	0.38355*** (4.92)	0.87533*** (7.18)	0.44248*** (5.67)	0.79797*** (7.29)
Prin4	-0.03267 (-0.42)	-0.23675** (-2.52)	0.00419 (0.05)	-0.43704*** (-2.84)	0.37785*** (4.86)	1.27134*** (4.03)
Prin5	0.17381** (2.23)	-0.18700 (-1.42)	0.38016*** (4.89)	-0.09309 (-0.44)	0.37619*** (4.82)	0.18198** (2.12)
Prin6	0.13128* (1.69)	0.14950 (1.41)	0.41432*** (5.31)	0.21610** (2.04)	-0.08175 (-1.05)	-1.28888*** (-5.84)
Prin7	0.02504 (0.32)	-1.10095*** (-7.39)	-0.14948* (-1.92)	-0.33366 (-1.18)	-0.16256** (-2.09)	0.55372** (2.31)
Prin8	-0.01394 (-0.18)	-1.20876*** (-7.54)	0.32636*** (4.19)	0.71336 (1.39)	0.27679*** (3.55)	1.73800*** (5.65)
Prin9			-0.34093*** (-4.37)	-0.54191*** (-3.99)	-0.39411*** (-5.05)	-0.31178*** (-2.77)
Prin10			0.28347*** (3.64)	0.36900*** (3.19)	0.37504*** (4.82)	-0.43263** (-2.34)
Prin11			-0.11199 (-1.44)	-0.03540 (-0.16)	-0.17480** (-2.25)	1.98082*** (4.51)
Prin12					0.20077** (2.57)	0.92494*** (4.11)
Prin13					0.20362*** (2.61)	0.35161*** (2.98)
Prin14					0.03437 (0.44)	0.66799*** (3.99)
Prin15					0.25376*** (3.23)	0.76721*** (5.83)
Prin16					-0.00981 (-0.13)	-0.69235*** (-4.48)
Year	No	Yes	No	Yes	No	Yes
Obs	313,016	313,016	313,016	313,016	313,016	313,016
F-statistic	25.83***	25.63***	28.51***	23.52***	22.18***	21.76***
Adj.R²	0.0007	0.0010	0.0011	0.0011	0.0011	0.0013

Notes: This table shows the relationship between trading duration and performance on account levels. The dependent variable is the account-level net cash flow per contract. *Duration* is the average time between trades of each account. *Prin1* through *Prin16* denote the factorized control variables using a principal component analysis method. Models *M2*, *M4*, and *M6* show the estimation results after controlling for the time effect (*Year*). The numbers in parentheses are the *t*-statistics. *Obs* denotes the number of observations. *F-statistic* shows the *F*-statistic value for model fitness. *Adj.R²* denotes the adjusted *R*-squared value. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 6

Correlation between the control variables and principal components

	Prin1	Prin2	Prin3	Prin4	Prin5	Prin6	Prin7	Prin8	Prin9	Prin10	Prin11	Prin12	Prin13	Prin14	Prin15	Prin16
Ret_Spot	0.972***	-0.165***	0.056***	0.041***	0.097***	0.038***	-0.045***	0.015***	-0.084***	0.025***	-0.031***	-0.002	-0.017***	-0.034***	-0.014***	0.000
Ret_Fu	0.976***	-0.136***	0.094***	0.041***	0.071***	0.047***	-0.014***	0.023***	-0.048***	0.040***	-0.033***	-0.059***	0.010***	0.035***	-0.013***	0.000
Cap_Spot	0.975***	-0.165***	0.047***	0.049**	0.053***	0.044***	-0.020***	0.020***	-0.087***	0.046***	-0.045***	-0.002	0.004**	-0.002	0.027***	0.000
Volat_Spot	-0.185***	0.973***	-0.007***	-0.010***	-0.074***	0.000	0.026***	-0.097***	0.010***	-0.038***	-0.024***	0.039***	-0.023***	-0.001	-0.005***	-0.004***
Volat_Fu	-0.214***	0.968***	0.001	-0.044***	-0.066***	0.006***	0.019***	-0.084***	0.019***	-0.033***	-0.022***	0.007***	-0.039***	0.001	0.005***	0.004***
Volat_Spot_r	-0.528***	0.488***	-0.327***	-0.053***	-0.086***	0.127***	0.088***	0.116***	0.241***	-0.158***	0.123***	0.480***	0.002	0.000	0.000	0.000
Volat_Fu_r	0.146***	-0.144***	-0.233***	-0.151***	0.004**	0.351***	0.288***	0.121***	-0.008***	0.811***	0.044***	-0.036***	-0.002	0.000	0.000	0.000
Volume_Spot	-0.079***	0.005***	-0.128***	0.166***	-0.054***	0.127***	0.949***	0.028***	-0.005***	0.172***	0.012***	0.015***	-0.002	0.000	0.000	0.000
Liq_Spr	-0.112***	0.747***	0.029***	0.301***	0.071***	0.052***	-0.122***	-0.516***	0.090***	-0.090***	-0.024***	0.028***	0.182***	0.001	0.000	0.000
Liq_Dep	0.041***	-0.490***	-0.129***	-0.253***	-0.179***	0.024***	0.020***	0.783***	-0.039***	0.113***	0.119***	0.040***	0.038***	0.000	0.000	0.000
Liq_Spr_r	-0.294***	-0.142***	-0.468***	-0.179***	-0.427***	-0.019***	0.036***	0.259***	-0.096***	0.096***	0.607***	0.062***	-0.001	0.000	0.000	0.000
Liq_Dep_r	0.132***	-0.012***	0.950***	0.049***	0.161***	0.003*	-0.129***	-0.058***	0.062***	-0.145***	-0.076***	-0.034***	0.000	0.000	0.000	0.000
Term_Spr_r	0.199***	-0.137***	0.215***	0.160***	0.868***	-0.110***	-0.066***	-0.125***	0.270***	0.012***	-0.101***	-0.015***	0.002	0.000	0.000	0.000
Credit_Spr_r	0.117***	0.045***	0.072***	0.918***	0.146***	0.054***	0.189***	-0.191***	-0.165***	-0.103***	-0.058***	-0.011***	0.001	0.000	0.000	0.000
RF_r	0.079***	0.041***	0.010***	0.056***	-0.082***	0.950***	0.123***	-0.001	-0.161***	0.196***	-0.006***	0.021***	0.001	0.000	0.000	0.000
Ex_Rate_r	-0.223***	0.075***	0.076***	-0.170***	0.245***	-0.189***	-0.006***	-0.042***	0.900***	-0.007***	-0.033***	0.037***	0.002	0.000	0.000	0.000

Notes: This table reports the correlations between the extracted principal components and the set of control variables in the spot and futures markets. Darker colors indicate a higher correlation. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 7

Impact of trading duration on investment performance by investor type

	Ind	Ins_I	Ins_NI	For_I	For_NI
Intercept	-4.42541*** (-4.15)	-6.45475*** (-3.38)	-11.12411 (-1.56)	-1.11935 (-0.28)	0.34763 (0.17)
Duration	-0.00002 (-0.47)	-0.00013* (-1.92)	0.00067** (2.53)	-0.00042*** (-2.74)	-0.00018** (-2.28)
Prin1	0.29431*** (3.34)	1.06720*** (7.14)	4.29747*** (7.56)	0.82962** (2.23)	0.71638*** (4.17)
Prin2	-2.54731*** (-3.14)	-4.17069*** (-2.87)	-4.77597 (-0.88)	-0.90061 (-0.29)	0.73045 (0.48)
Prin3	0.42011*** (3.41)	1.23544*** (5.58)	3.41744*** (4.13)	0.67849 (1.44)	0.25052 (1.09)
Prin4	1.23214*** (3.47)	1.66490*** (2.60)	2.49355 (1.04)	0.22564 (0.17)	-0.15249 (-0.23)
Prin5	-0.08680 (-0.90)	0.40561** (2.42)	1.47053** (2.34)	0.76069* (1.93)	0.11924 (0.64)
Prin6	-0.65903*** (-2.65)	-2.21650*** (-5.14)	-5.11710*** (-3.11)	-0.84893 (-0.86)	-0.24622 (-0.51)
Prin7	0.91862*** (3.40)	0.54824 (1.10)	-0.94967 (-0.52)	0.52745 (0.54)	-0.17269 (-0.35)
Prin8	1.41722*** (4.09)	2.62751*** (4.33)	4.66640** (2.04)	0.24569 (0.18)	0.42209 (0.64)
Prin9	0.04020 (0.32)	-0.29206 (-1.30)	-4.35940*** (-5.10)	0.88203* (1.73)	-0.21027 (-0.87)
Prin10	-0.39689* (-1.91)	-0.97100*** (-2.59)	0.64854 (0.46)	-0.07167 (-0.09)	0.16452 (0.42)
Prin11	1.85218*** (3.75)	3.19834*** (3.62)	3.64419 (1.10)	1.54030 (0.82)	-0.30824 (-0.33)
Prin12	1.17482*** (4.63)	1.12222** (2.46)	0.66314 (0.39)	0.79554 (0.83)	-0.02651 (-0.06)
Prin13	0.39820*** (3.00)	0.48572** (2.00)	0.04245 (0.05)	0.37302 (0.77)	-0.04075 (-0.17)
Prin14	0.51343*** (2.72)	0.88598*** (2.63)	1.61534 (1.29)	-0.32151 (-0.45)	0.27848 (0.78)
Prin15	0.47668*** (3.22)	1.22504*** (4.71)	4.66585*** (4.73)	-0.83927 (-1.44)	-0.06536 (-0.23)
Prin16	-0.47814*** (-2.74)	-1.19854*** (-3.89)	-1.90511* (-1.67)	-0.78294 (-1.16)	0.18437 (0.56)
Year	Yes	Yes	Yes	Yes	Yes
Obs	136,737	69,798	23,424	22,329	60,256
F-statistic	6.42***	10.34***	12.45***	2.62***	2.82***
Adj.R²	0.0008	0.0025	0.0092	0.0014	0.0006

Notes: This table shows the relationship between trading duration and investor performance. The dependent variable is the account-level net cash flow per contract. *Duration* is the average time between trades of each account. *Prin1* through *Prin16* denote the factorized control variables using a principal component analysis method. The columns labeled *Ind*, *Ins_I*, *Ins_NI*, *For_I*, and *For_NI* show the subsample analyses for domestic individuals, domestic investment firms, domestic non-investment institutions, foreign investment firms, and foreign non-investment institutions, respectively. The numbers in parentheses are the *t*-statistics. The row labeled *Year* shows whether the model controls for the year effect, and that labeled *Obs* shows the number of observations used for estimation. *F-statistic* shows the *F*-statistic value for model fitness. *Adj.R²* denotes the adjusted *R*-squared value. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 8

Relationship between trading duration and investment performance throughout the day

	Ind	Ins_I	Ins_NI	For_I	For_NI
09:00-10:00	0.00028 (1.49)	-0.00027 (-0.88)	0.00313** (2.36)	0.00049 (0.62)	0.00084** (2.32)
10:00-11:00	0.00010* (1.73)	-0.00015 (-1.35)	-0.00008 (-0.17)	-0.00026 (-1.02)	-0.00012 (-0.94)
11:00-12:00	0.00000 (0.00)	-0.00013* (-1.87)	-0.00001 (-0.05)	-0.00029* (-1.82)	-0.00013 (-1.63)
12:00-13:00	-0.00000 (-0.16)	0.00005 (1.03)	0.00001 (0.03)	-0.00008 (-0.74)	-0.00006 (-1.13)
13:00-14:00	-0.00005** (-2.31)	-0.00004 (-1.01)	0.00011 (0.67)	-0.00015 (-1.57)	-0.00017*** (-3.80)
14:00-15:00	-0.00001 (-0.61)	-0.00006* (-1.65)	0.00006 (0.43)	-0.00033*** (-4.17)	-0.00018*** (-4.35)

Notes: This table shows the relationship between investors' trading duration and performance depending on their types and the time of the day. The columns labeled *Ind*, *Ins_I*, *Ins_NI*, *For_I*, and *For_NI* show the subsample analyses for domestic individuals, domestic investment firms, domestic non-investment institutions, foreign investment firms, and foreign non-investment institutions, respectively. The leftmost column shows the hourly intraday intervals from 9:00 to 15:00. The numbers in parentheses are the *t*-statistics. The estimation models in this table all consider the year effect, and control variables include the year effect 16 principal components extracted from underlying and futures market fundamentals and macroeconomic fluctuations. The estimation results for control variables are omitted for brevity. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 9

Relationship between trading duration and investment performance by liquidity state

Panel A. During high liquidity periods

	Ind	Ins_I	Ins_NI	For_I	For_NI
Sprd(B30)	-0.00010 (-1.63)	-0.00028** (-2.20)	0.00065 (1.45)	-0.00031 (-1.18)	0.00012 (0.87)
Dep(T30)	-0.00004 (-0.62)	-0.00018 (-1.22)	0.00038 (0.78)	-0.00045 (-1.52)	0.00016 (1.08)

Panel B. During low liquidity periods

	Ind	Ins_I	Ins_NI	For_I	For_NI
Sprd(T30)	-0.00005 (-0.81)	0.00002 (0.19)	0.00131*** (3.02)	0.00004 (0.14)	-0.00032** (-2.47)
Dep(B30)	0.00001 (0.07)	0.00018 (1.42)	0.00249*** (4.80)	-0.00065* (-1.95)	-0.00032** (-2.05)

Notes: This table shows the relationship between investors' trading duration and performance depending on their types and market liquidity states. Panels A and B show the results during the high and low liquidity periods, respectively. The columns labeled *Ind*, *Ins_I*, *Ins_NI*, *For_I*, and *For_NI* show the subsample analyses for domestic individuals, domestic investment firms, domestic non-investment institutions, foreign investment firms, and foreign non-investment institutions, respectively. *Sprd(B30)* and *Sprd(T30)* denote the periods with the bottom 30-percentile and top 30-percentile bid-ask spread values, respectively; *Dep(B30)* and *Dep(T30)* denote the periods with the bottom 30-percentile and top 30-percentile market depth values, respectively. The numbers in parentheses are the *t*-statistics. The estimation models in this table all consider the year effect, and control variables include the year effect 16 principal components extracted from underlying and futures market fundamentals and macroeconomic fluctuations. The estimation results for control variables are omitted for brevity. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 10

Relationship between duration and performance by duration quintile

Panel A. Investor composition in each quintile of trading duration

	Q5	Q4	Q3	Q2	Q1
Ind	46.36%	46.13%	45.97%	42.89%	38.74%
Ins_I	22.18%	23.24%	23.24%	24.12%	20.16%
Ins_NI	4.18%	2.84%	2.88%	6.62%	17.43%
For_I	7.26%	7.45%	7.27%	6.96%	6.82%
For_NI	20.02%	20.34%	20.64%	19.41%	16.84%

Panel B. Impact of trading duration on investment performance in each quintile

	Q5	Q4	Q3	Q2	Q1
Intercept	-4.12169** (-2.33)	-0.11383 (-0.08)	-1.88000 (-1.26)	-2.76768 (-1.60)	-14.33462*** (-3.84)
Duration	-0.00005 (-0.98)	-0.00007 (-0.22)	-0.00007 (-0.12)	-0.00056 (-0.74)	0.00455** (2.37)
Prin1	0.80053*** (5.24)	0.49018*** (4.43)	0.36011*** (3.35)	0.61908*** (4.80)	1.90868*** (6.38)
Prin2	-1.95406 (-1.46)	0.63214 (0.64)	-0.77118 (-0.78)	-1.74229 (-1.41)	-9.16988*** (-3.22)
Prin3	0.29601 (1.48)	0.12355 (0.83)	0.49158*** (3.26)	0.45685** (2.36)	2.39089*** (5.54)
Prin4	1.01705* (1.75)	0.02768 (0.06)	0.49603 (1.15)	0.70457 (1.28)	3.95506*** (3.19)
Prin5	-0.37794** (-2.29)	-0.00900 (-0.07)	0.17831 (1.52)	0.22395 (1.56)	0.57875* (1.75)
Prin6	-0.82420** (-2.01)	0.05439 (0.18)	-0.45603 (-1.52)	-1.02272*** (-2.74)	-4.05372*** (-4.66)
Prin7	0.86490** (1.99)	0.11069 (0.34)	0.44994 (1.37)	0.28716 (0.68)	1.05670 (1.11)
Prin8	1.68586*** (2.95)	0.04011 (0.10)	0.61679 (1.47)	1.23136** (2.35)	5.15432*** (4.25)
Prin9	-0.23298 (-1.13)	-0.37593** (-2.46)	-0.22009 (-1.42)	-0.06836 (-0.34)	-0.95188** (-2.14)
Prin10	-0.31139 (-0.92)	0.50240** (2.00)	-0.29543 (-1.17)	-0.41320 (-1.29)	-1.69048** (-2.30)
Prin11	1.66078** (2.04)	-0.49824 (-0.83)	0.60783 (1.01)	1.42360* (1.90)	6.69993*** (3.86)
Prin12	1.52329*** (3.64)	0.59635* (1.94)	0.52730* (1.72)	0.49596 (1.29)	1.46234* (1.65)
Prin13	0.49381** (2.28)	0.25928 (1.62)	0.02207 (0.14)	0.16950 (0.82)	0.65138 (1.42)
Prin14	0.45273 (1.47)	0.02363 (0.10)	0.54225** (2.36)	0.27082 (0.94)	2.14019*** (3.25)
Prin15	0.05344 (0.22)	-0.08734 (-0.49)	0.14968 (0.83)	0.86460*** (3.80)	2.97862*** (5.69)
Prin16	-0.52767* (-1.80)	0.13679 (0.64)	-0.40884* (-1.93)	-0.19069 (-0.73)	-2.25157*** (-3.78)
Year	Yes	Yes	Yes	Yes	Yes
Obs	62,582	62,611	62,616	62,601	62,606
F-statistic	7.57***	6.17***	4.40***	4.37***	11.24***

Adj.R^2	0.0020	0.0016	0.0010	0.0010	0.0031
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Notes: This table shows the relationship between trading duration and investor performance by duration quintile. Panel A shows the proportion of each investor type in each quintile of trading duration. *Ind*, *Ins_I*, *Ins_NI*, *For_I*, and *For_NI* denote domestic individuals, domestic investment firms, domestic non-investment institutions, foreign investment firms, and foreign non-investment institutions, respectively. *Q1* (*Q5*) denotes the quintile with the lowest (highest) trading duration. Panel B shows the estimation results for each quintile. *Duration* denotes trading duration. *Prin1* through *Prin16* denote the factorized control variables using a principal component analysis method. The numbers in parentheses are the *t*-statistics. *Year* indicates that the model controls for the year effect, and that labeled *Obs* shows the number of observations used for estimation. *F-statistic* shows the *F*-statistic value for model fitness. *Adj. R^2* denotes the adjusted *R*-squared value. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 11

Relationship between duration and performance by market state

Panel A. Descriptive statistics of trading duration by investor type in bull and bear markets

	Ind		Ins_I		Ins_NI		For_I		For_NI	
	Bear	Bull	Bear	Bull	Bear	Bull	Bear	Bull	Bear	Bull
Pct15	2,094 (2,378)	2,081 (2,290)	1,923 (2,229)	1,889 (2,149)	1,409 (2,264)	1,405 (2,208)	1,952 (2,460)	1,755 (2,131)	1,893 (2,207)	1,813 (2,149)
Pct30	2,195 (2,438)	2,125 (2,340)	1,987 (2,216)	1,911 (2,126)	1,444 (2,174)	1,394 (2,091)	1,943 (2,334)	1,830 (2,238)	1,928 (2,161)	1,854 (2,108)

Panel B. Mean performance by investor type in bull and bear markets

	Ind		Ins_I		Ins_NI		For_I		For_NI	
	Bear	Bull	Bear	Bull	Bear	Bull	Bear	Bull	Bear	Bull
Pct15	-1.222	0.925	-1.569	1.446	-8.532	3.475	-2.687	2.178	-1.744	1.081
Pct30	-1.164	0.455	-1.459	1.142	-7.595	5.811	-2.099	-0.500	-1.211	1.096

Notes: This table shows relationship between duration and performance by market state. Panel A shows mean and standard deviation values of investors' trading duration in bear (*Bear*) and bull (*Bull*) markets. The columns labeled *Ind*, *Ins_I*, *Ins_NI*, *For_I*, and *For_NI* denote domestic individuals, domestic investment firms, domestic non-investment institutions, foreign investment firms, and foreign non-investment institutions, respectively. The rows labeled *Pct15* (*Pct30*) denote the bull and bear markets, which are identified as periods with top and bottom 15% (30%) values of returns, respectively. Panel B shows mean values of investors' performances in bear and bull markets.

Table 12

Impact of trading duration on investor performance by investor type after adjusting for the price level

	Ind	Ins_I	Ins_NI	For_I	For_NI
Intercept	-0.0174*** (-4.07)	-0.0265*** (-3.45)	-0.0489* (-1.70)	-0.0030 (-0.19)	0.0003 (0.04)
Duration	-0.0000 (-0.54)	-0.0000* (-1.95)	0.0000** (2.44)	-0.0000*** (-2.72)	-0.0000** (-2.28)
Prin1	0.0012*** (3.40)	0.0043*** (7.16)	0.0171*** (7.46)	0.0033** (2.23)	0.0028*** (4.08)
Prin2	-0.0100*** (-3.08)	-0.0174*** (-2.99)	-0.0233 (-1.06)	-0.0025 (-0.20)	0.0019 (0.31)
Prin3	0.0016*** (3.29)	0.0049*** (5.54)	0.0137*** (4.11)	0.0025 (1.33)	0.0010 (1.11)
Prin4	0.0049*** (3.44)	0.0071*** (2.75)	0.0124 (1.29)	0.0002 (0.04)	-0.0002 (-0.08)
Prin5	-0.0003 (-0.87)	0.0016** (2.39)	0.0057** (2.25)	0.0033** (2.10)	0.0004 (0.57)
Prin6	-0.0025** (-2.51)	-0.0088*** (-5.06)	-0.0200*** (-3.02)	-0.0033 (-0.84)	-0.0012 (-0.61)
Prin7	0.0035*** (3.23)	0.0023 (1.14)	-0.0029 (-0.38)	0.0019 (0.47)	-0.0005 (-0.25)
Prin8	0.0055*** (3.96)	0.0107*** (4.38)	0.0199** (2.16)	0.0003 (0.05)	0.0021 (0.79)
Prin9	0.0001 (0.29)	-0.0011 (-1.18)	-0.0175*** (-5.06)	0.0037* (1.80)	-0.0007 (-0.72)
Prin10	-0.0017** (-1.99)	-0.0041*** (-2.74)	0.0011 (0.18)	0.0000 (0.00)	0.0005 (0.29)
Prin11	0.0073*** (3.67)	0.0131*** (3.67)	0.0165 (1.23)	0.0055 (0.73)	-0.0006 (-0.16)
Prin12	0.0046*** (4.58)	0.0047** (2.54)	0.0035 (0.52)	0.0030 (0.77)	0.0001 (0.07)
Prin13	0.0016*** (2.97)	0.0020** (2.05)	0.0008 (0.23)	0.0013 (0.69)	-0.0001 (-0.12)
Prin14	0.0021*** (2.71)	0.0037*** (2.73)	0.0074 (1.46)	-0.0015 (-0.53)	0.0013 (0.91)
Prin15	0.0019*** (3.21)	0.0051*** (4.83)	0.0196*** (4.93)	-0.0037 (-1.57)	-0.0001 (-0.07)
Prin16	-0.0019*** (-2.65)	-0.0049*** (-3.95)	-0.0086* (-1.85)	-0.0029 (-1.06)	0.0006 (0.45)
Year	Yes	Yes	Yes	Yes	Yes
Obs	136,737	69,798	23,424	22,329	60,256
F-statistic	6.16***	10.39***	12.45***	2.71***	2.70
Adj.R²	0.0007	0.0025	0.0092	0.0015	0.0005

Notes: This table shows the relationship between trading duration and performance on account levels. The dependent variable is the account-level net cash flow per contract. *Duration* is the average time between trades of each account. *Prin1* through *Prin16* denote the factorized control variables using a principal component analysis method. The columns labeled *Ind*, *Ins_I*, *Ins_NI*, *For_I*, and *For_NI* show the subsample analyses for domestic individuals, domestic investment firms, domestic non-investment institutions, foreign investment firms, and foreign non-investment institutions, respectively. The numbers in parentheses are the *t*-statistics. The row labeled *Year* shows whether the model controls for the year effect, and that labeled *Obs* shows the number of observations used for estimation. *F-statistic* shows the *F*-statistic value for model fitness. *Adj.R²* denotes the adjusted *R*-squared value. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 13

Robustness test using Fama-MacBeth regressions

Panel A. On a yearly basis

	Ind	Ins_I	Ins_NI	For_I	For_NI
Intercept	0.2932 (0.62)	0.9495 (2.02)	1.0145 (0.25)	1.2248 (1.77)	-1.8571 (-1.85)
Duration	-0.0000 (-0.24)	-0.0001 (-0.81)	0.0008** (3.33)	-0.0003* (-2.30)	-0.0001* (-2.30)

Panel B. On an expiration-to-expiration basis

	Ind	Ins_I	Ins_NI	For_I	For_NI
Intercept	-0.6146* (-2.01)	-0.5501 (-0.85)	-4.1591* (-1.80)	0.0687 (0.13)	-0.1498 (-0.43)
Duration	-0.0000 (-0.18)	-0.0001 (-0.89)	0.0008** (2.20)	-0.0003 (-1.46)	-0.0002** (-2.12)

Notes: This table shows the relationship between investors' performances and trading durations using Fama-MacBeth regressions. The columns labeled *Ind*, *Ins_I*, *Ins_NI*, *For_I*, and *For_NI* show the subsample analyses for domestic individuals, domestic investment firms, domestic non-investment institutions, foreign investment firms, and foreign non-investment institutions, respectively. The estimation results for control variables are omitted for brevity. ** and * indicate statistical significance at the 5% and 10% levels, respectively.